Electrical Circuits II
(EE-1201)

Lecture-03

Network Theorems

(Thevenin's Theorem – *cont..*)

Dr. M. N. Alam, EED, NITW

Outlines

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This unit covers:

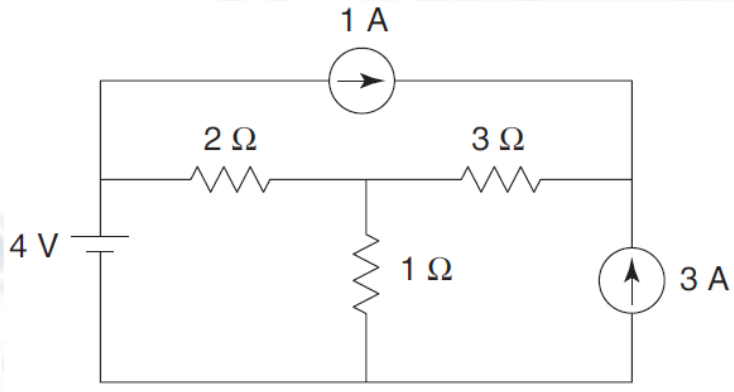
- Thevenin's theorem
- More numerical examples
- Homework

Thevenin's Theorem

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Numerical Example-8:

Determine the current through the 1 Ω resistor using Thevenin's theorem.

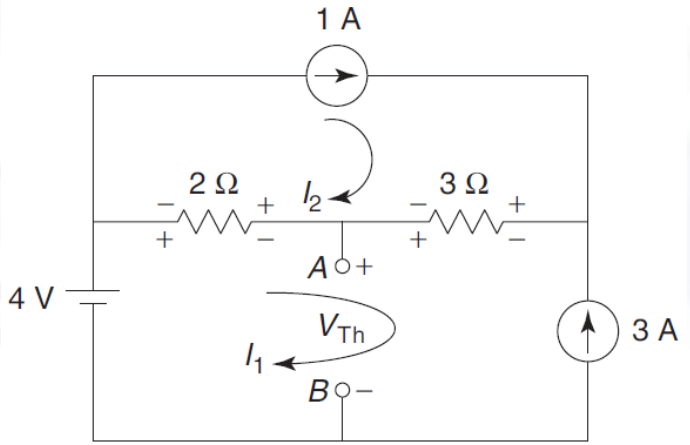


Solution:

Step I: Calculation of V_{Th},

- Writing the current equations for Meshes 1 and 2,

$$I_1 = -3\ A$$
$$I_2 = 1\ A$$



Thevenin's Theorem

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- Writing voltage equation,

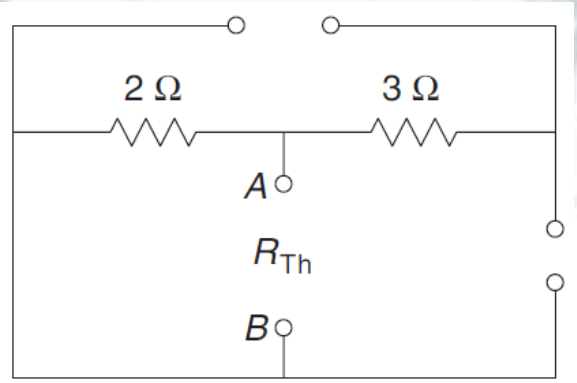
$$4 - 2(I_1 - I_2) - V_{Th} = 0$$

$$V_{Th} = 4 - 2(I_1 - I_2)$$

$$= 4 - 2 \times (-3 - 1) = 12\ V$$

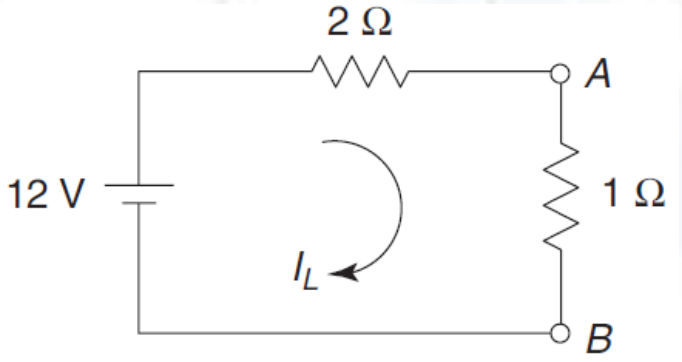
Step II: Calculation of R_{Th},

$$R_{Th} = 2\ \Omega$$



Step III: Calculation of I_L,

$$I_L = \frac{12}{2+1} = 4\ A$$

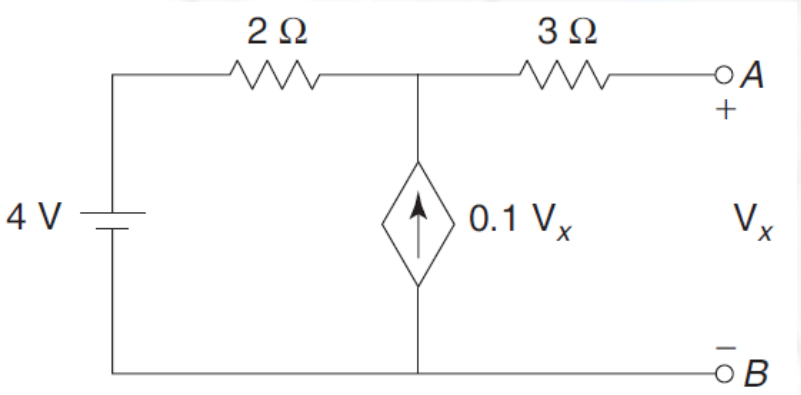


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Numerical Example-9:

Find the Thevenin's equivalent network for the circuit shown below.

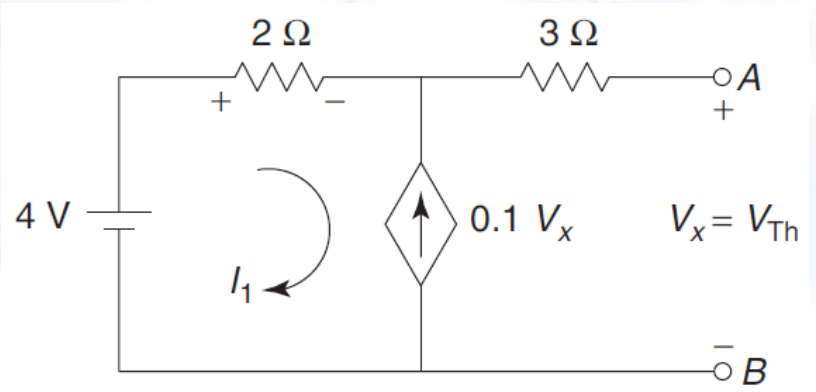


Solution:

Step I: Calculation of V_{Th},

- From the network;

$$V_{Th} = V_x$$
$$I_1 = -0.1\,V_x$$



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- Writing voltage equation,

$$4 - 2I_1 - V_x = 0$$
$$4 - 2(-0.1V_x) - V_x = 0$$
$$4 - 0.8V_x = 0$$
$$V_x = 5\ V$$
$$V_{Th} = V_x = 5\ V$$

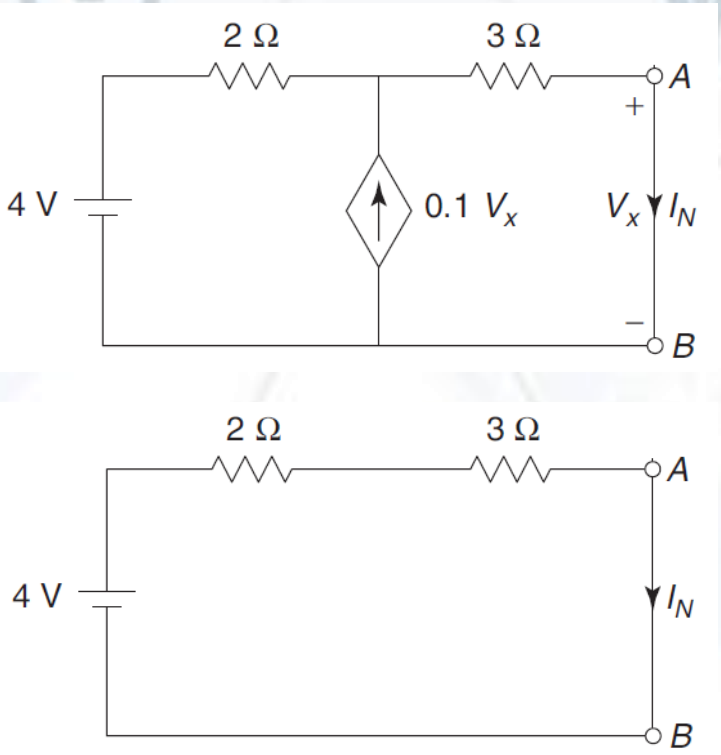
Step II: Calculation of I_N,

- As A-B terminal is short-circuited,

$$V_x = 0$$

- So, the dependent source, $0.1V_x = 0$ and network gets simplified as shown,
- Hence,

$$I_N = \frac{4}{2+3} = 0.8\ A$$



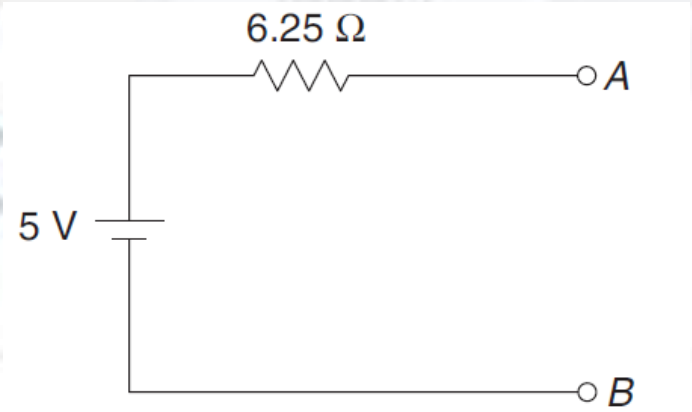
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Step III: Calculation of R_{Th},

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{5}{0.8} = 6.25\ \Omega$$

Step IV: Thevenin's equivalent network,

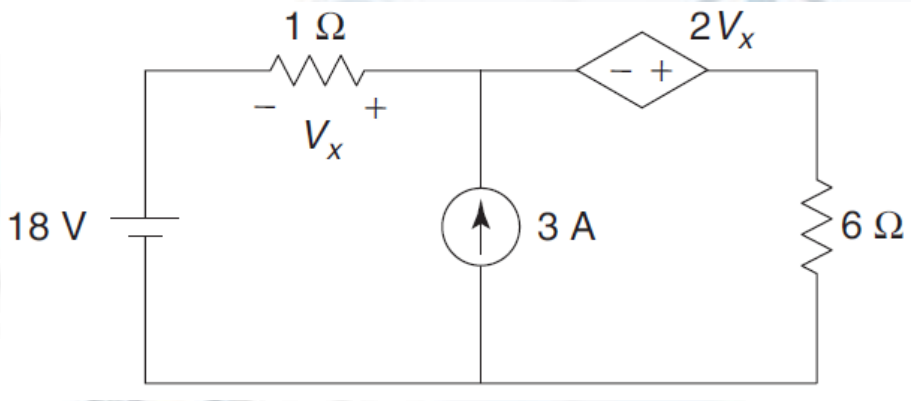


Thevenin's Theorem

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Numerical Example-10:

Find current through 6 Ω resistor using Thevenin's equivalent for the circuit shown.

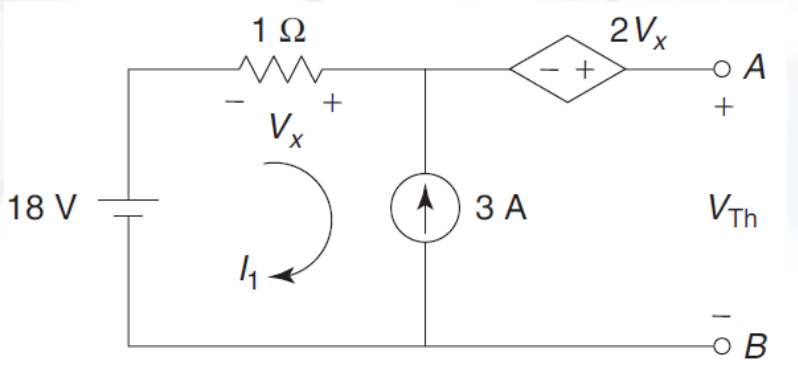


Solution:

Step I: Calculation of V_{Th},

- From the network;

$$I_1 = -3\ A$$
$$V_x = -1I_1 = 3\ V$$



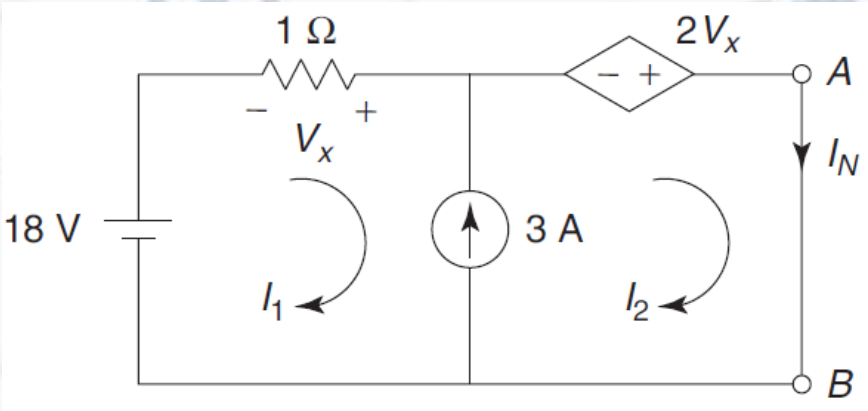
Thevenin's Theorem

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- Writing voltage equation,

$$18 + V_x + 2V_x - V_{Th} = 0$$
$$18 + (3) + 2(3) - V_{Th} = 0$$
$$V_{Th} = 27\ V$$

Step II: Calculation of I_N,



- From mesh 1,

$$V_x = -I_1 \ldots . (i)$$

- Meshes 1 and 2 will form a supermesh,

$$I_2 - I_1 = 3 \ldots . (ii)$$

- Applying KVL to the outer path of the supermesh,

$$18 - 1I_1 + 2V_x = 0$$
$$18 - 1I_1 + 2(-I_1) = 0$$
$$I_1 = 6\ A \ldots (iii)$$

- Solving Eqs (ii) and (iii),

$$I_2 = 9\ A \Rightarrow I_N = 9\ A$$

Thevenin's Theorem

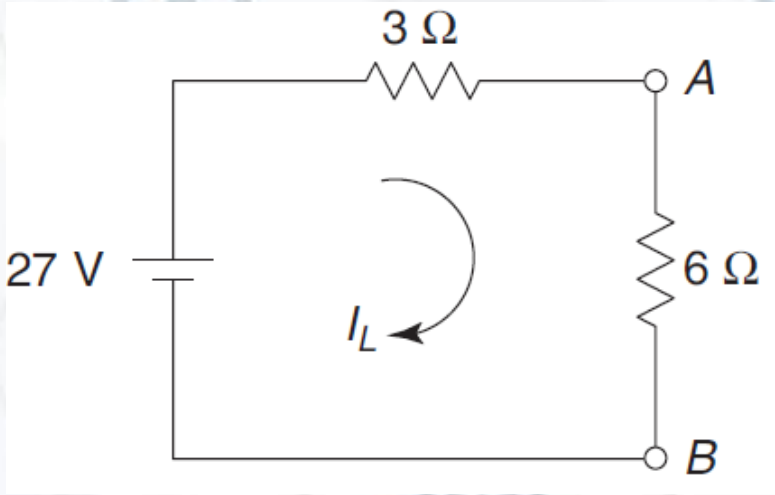
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Step III: Calculation of R_{Th},

$$R_{Th} = \frac{V_{Th}}{I_N} = \frac{27}{9} = 3\ \Omega$$

Step IV: Calculation of I_L,

$$I_L = \frac{27}{3+6} = 3\ A\ (\downarrow)$$

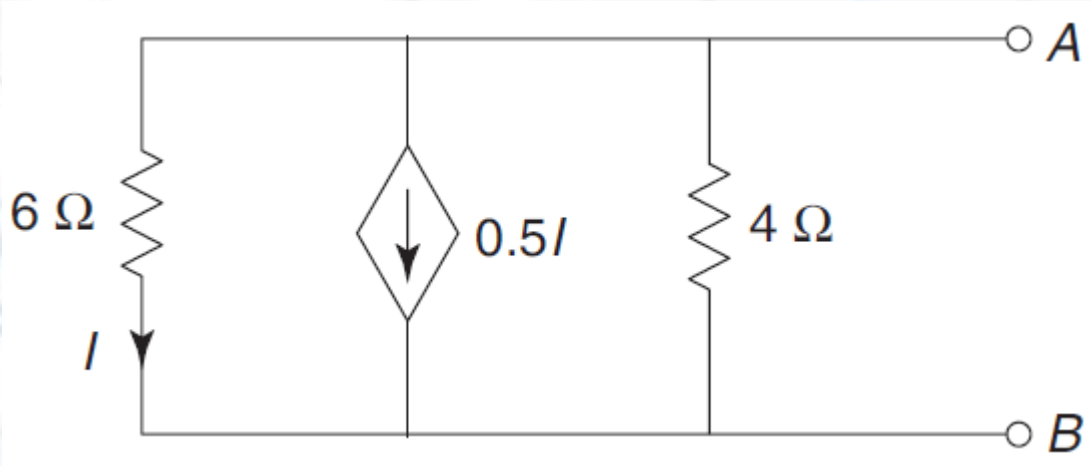


Thevenin's Theorem

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Numerical Example-11:

Find Thevenin's equivalent to the left of the terminal A-B.



Solution:

Step I: Calculation of V_{Th}

Since the network does not contain any independent source,

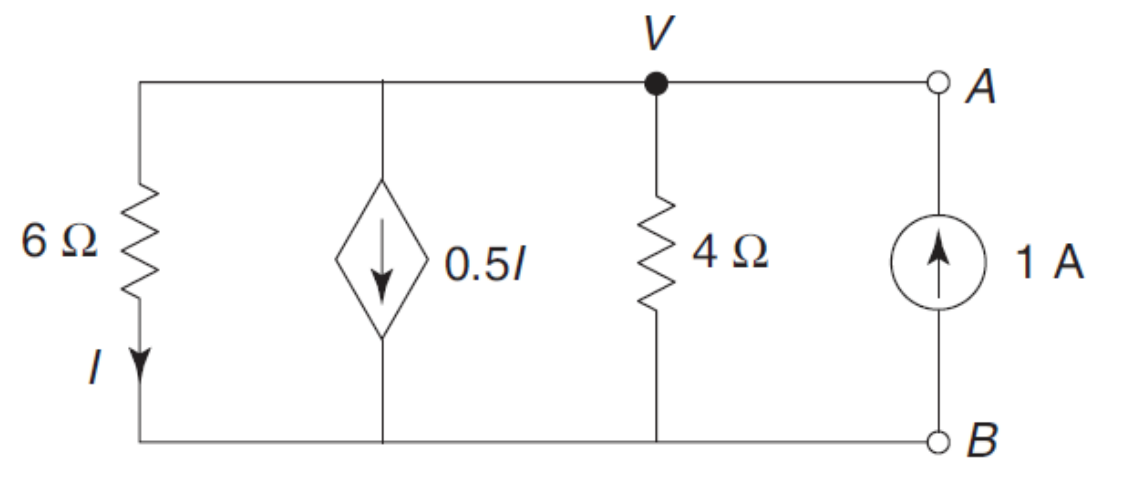
$$V_{Th} = 0$$
$$I_N = 0$$

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Step II: Calculation of R_{Th},

- R_{Th} can be calculated by applying a known current source of 1 A at the terminals A and B as shown in figure. Also, assume that terminal A voltage is V (unknown)



- We can write that ,

$$I = \frac{V}{6}$$

- Applying KCL at the node,

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Step II: Calculation of R_{Th},

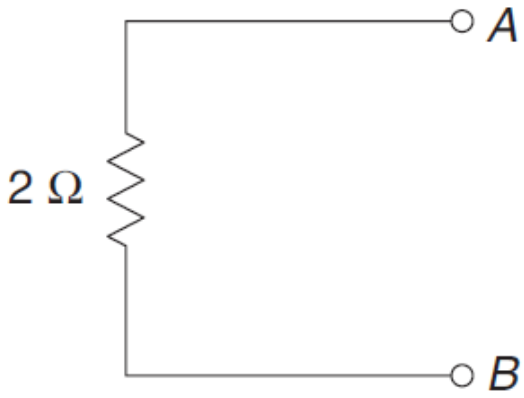
$$\frac{V}{6} + 0.5I + \frac{V}{4} = 1$$

$$\Rightarrow \quad \frac{V}{6} + 0.5 \times \left(\frac{V}{6}\right) + \frac{V}{4} = 1$$

$$\Rightarrow \quad \left(\frac{1}{6} + \frac{0.5}{6} + \frac{1}{4}\right) V = 1$$

$$V = 2$$

$$R_{Th} = \frac{V}{I} = \frac{2}{1} = 2\ \Omega$$



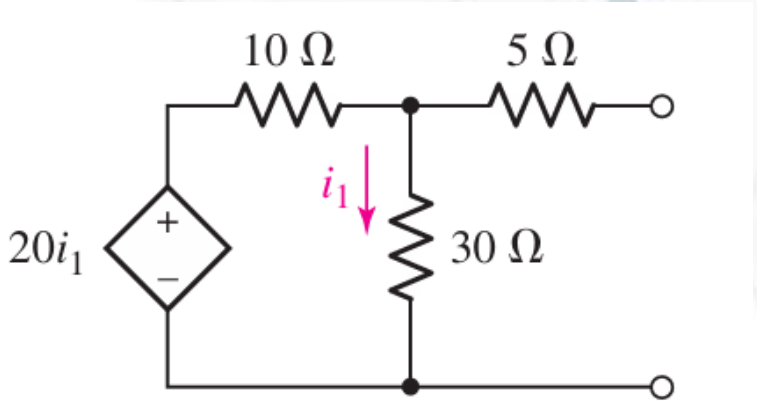
Step III: Thevenin's equivalent

Thevenin's Theorem

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Homework #2:

Find the Thevenin's equivalent of the network shown below.



Answer:

$$V_{Th} = 0; \;\; R_{Th} = 20$$